EASY · ALGEBRA · ANSWER KEY

Algebra

26

8 answers · Rationale-rich · Teach from doc

Q1

D

D. 55

Choice D is correct. In the given equation, s is the speed, in miles per hour, of a certain car t seconds after it began to accelerate. Therefore, the speed of the car, in miles per hour, 5 seconds after it began to accelerate can be found by substituting 5 for t in the given equation, which yields $s = 40 + 35$, or $s = 55$. Thus, the speed of the car 5 seconds after it began to accelerate is 55 miles per hour.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q2

D

D. 88

Choice D is correct. The value of gx when $x = 8$ can be found by substituting 8 for x in the given equation $gx = 10x + 8$. This yields $g8 = 108 + 8$, or $g8 = 88$. Therefore, when $x = 8$, the value of gx is 88.

Choice A is incorrect. This is the value of x when $gx = 8$, rather than the value of gx when $x = 8$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q3

2

The correct answer is 2. Substituting 8 for fx in the given equation yields $8 = 4x$. Dividing the left- and right-hand sides of this equation by 4 yields $x = 2$. Therefore, the value of x is 2 when $fx = 8$.

Q4

C

C. $\$ 15{,}400$

Choice C is correct. It's given that fx is the total cost, in dollars, to lease a car from this dealership with a monthly payment of x dollars. Therefore, the total cost, in dollars, to lease the car when the monthly payment is $\$ 400$ is represented by the value of fx when $x = 400$. Substituting 400 for x in the equation $fx = 36x + 1{,}000$ yields $f400 = 36400 + 1{,}000$, or $f400 = 15{,}400$. Thus, when the monthly payment is $\$ 400$, the total cost to lease a car is $\$ 15{,}400$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q5

C

C. 3

Choice C is correct. If $f(x) = \frac{2x-1}{3}$, then $f(5) = \frac{2(5)-1}{3} = \frac{10-1}{3} = \frac{9}{3} = 3$.

Choice A is incorrect and may result from not multiplying x by 2 in the numerator. Choice B is incorrect and may result from dividing 2x by 3 and then subtracting 1. Choice D is incorrect and may result from evaluating only the numerator 2x – 1.

Q6

A

A. $h(x) = -x + 41$

Choice A is correct. An equation defining a linear function can be written in the form $hx = ax + b$, where a and b are constants. It's given that $h0 = 41$. Substituting 0 for x and 41 for hx in the equation $hx = ax + b$ yields $41 = a0 + b$, or $b = 41$. Substituting 41 for b in the equation $hx = ax + b$ yields $hx = ax + 41$. It's also given that $h1 = 40$. Substituting 1 for x and 40 for hx in the equation $hx = ax + 41$ yields $40 = a1 + 41$, or $40 = a + 41$. Subtracting 41 from the left- and right-hand sides of this equation yields $-1 = a$. Substituting -1 for a in the equation $hx = ax + 41$ yields $hx = -1x + 41$, or $hx = -x + 41$.

Choice B is incorrect. Substituting 0 for x and 41 for hx in this equation yields $41 = -0$, which isn't a true statement.

Choice C is incorrect. Substituting 0 for x and 41 for hx in this equation yields $41 = -410$, or $41 = 0$, which isn't a true statement.

Choice D is incorrect. Substituting 41 for hx in this equation yields $41 = -41$, which isn't a true statement.

Q7

D

D. $f(x) = 39x$

Choice D is correct. An equation defining a linear function can be written in the form $fx = mx + b$, where m is the slope and $0, b$ is the y-intercept of the graph of $y = fx$ in the xy-plane. It's given that the graph of $y = fx$ has a slope of 39, so $m = 39$. It's also given that the graph of $y = fx$ passes through the point $0, 0$, so $b = 0$. Substituting 39 for m and 0 for b in $fx = mx + b$ yields $fx = 39x + 0$, or $fx = 39x$. Thus, the equation that defines f is $fx = 39x$.

Choice A is incorrect. This equation defines a function whose graph has a slope of -39, not 39.

Choice B is incorrect. This equation defines a function whose graph has a slope of $\frac{1}{39}$, not 39.

Choice C is incorrect. This equation defines a function whose graph has a slope of 1, not 39, and passes through the point $0, -39$, not $0, 0$.

Q8 C

C. $fx = 60x + 120$

Choice C is correct. It's given that the technician charges $\$\ 60$ per hour for labor. Therefore, if the job takes x hours, the technician will charge $\frac{\$\ 60}{1 \text{ hour}}x$ hours, or $\$\ 60x$, for labor. It's also given that the technician charges $\$\ 120$ for parts. Therefore, $fx = 60x + 120$ represents the total amount, in dollars, the technician will charge for this job if it takes x hours.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This function represents the total amount, in dollars, the technician charges for labor only, not the total amount charged for labor and parts.

Choice D is incorrect. This function represents the total amount, in dollars, the technician would charge if the charge for parts were subtracted from, rather than added to, the charge for labor.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

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